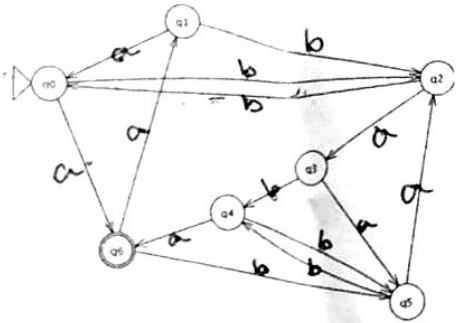
B. Tech. (Computer Engineering) Vth Semester
Ist Sessional, AUTOMATA THEORY (CSC-502)

Time: 1 Hour **MM:12**

Note: Attempt all questions. **Date: 14/09/26**

Your Roll No : .....

Q No.	CO Mapped
1	1
2	1
3	2
4	2

Q. No	Question	Marks	BL
1.	Design a Mealy as well as a Moore machine that takes an input string $w \in \{a, b\}^*$ and outputs **1** whenever the input string ends with the sequence aba or bab, else outputs **0** .	3	6
2.	Minimise the following DFA :  ***Verify your answer using suitable examples.***	3	4
3.	Design a Regular Expression for language L_1 defined over the alphabet {0, 1} such that L_1 contains all strings where the number of **0**s are odd and does not contain the substring '***100***'. ***Verify your answer using suitable examples.***	3	3
4	Comment on the Regularity of the following language which generate strings as follows : **$L = \{ a^i b^j c^k \mid i, j, k >= 0$ & if i=1 then j=k $\}$**	3	4